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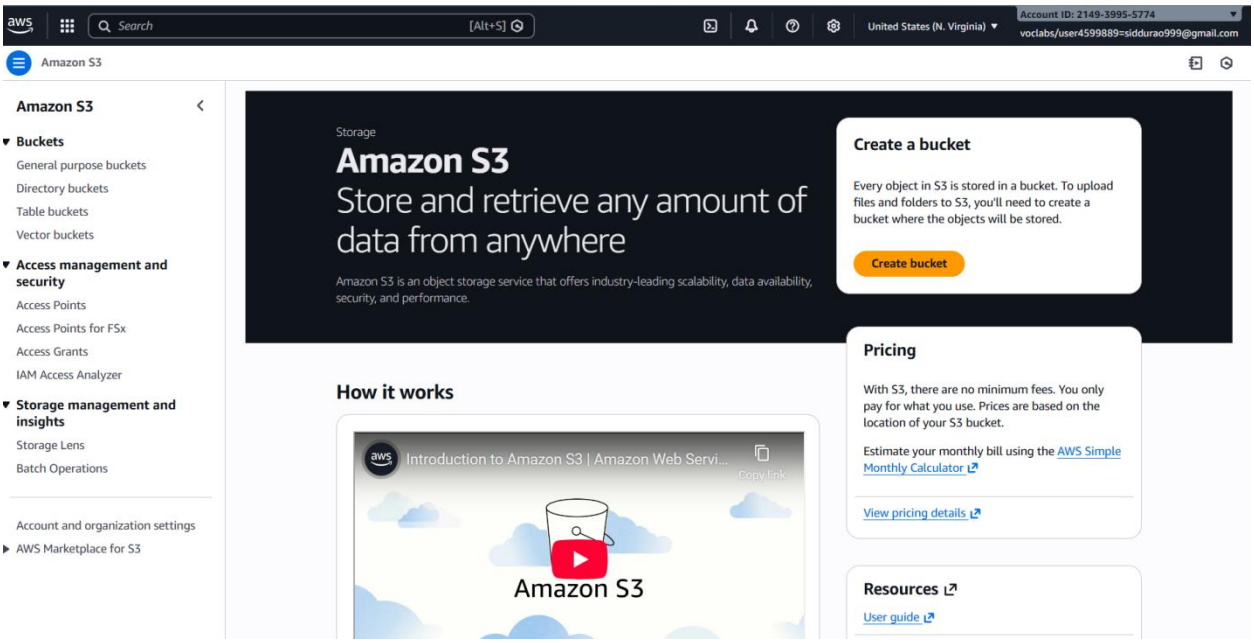
CC_LAB_WEEK4

1.Creating an S3 bucket, upload an object, and enable public access

1⃣Create an S3 Bucket (with Public Access Enabled)

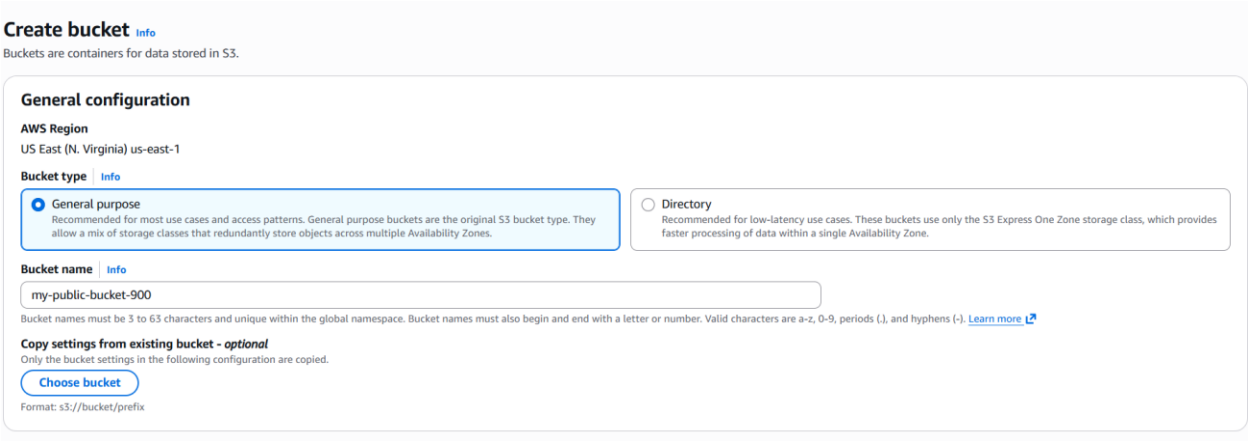
Step 1: Open S3 Service

- Login to AWS Management Console
- Go to Services → S3
- Click Create bucket



Step 2: Bucket Configuration

- Bucket name: my-public-bucket-123
- Region: Choose nearest region



Step 3: Disable Block Public Access (IMPORTANT)

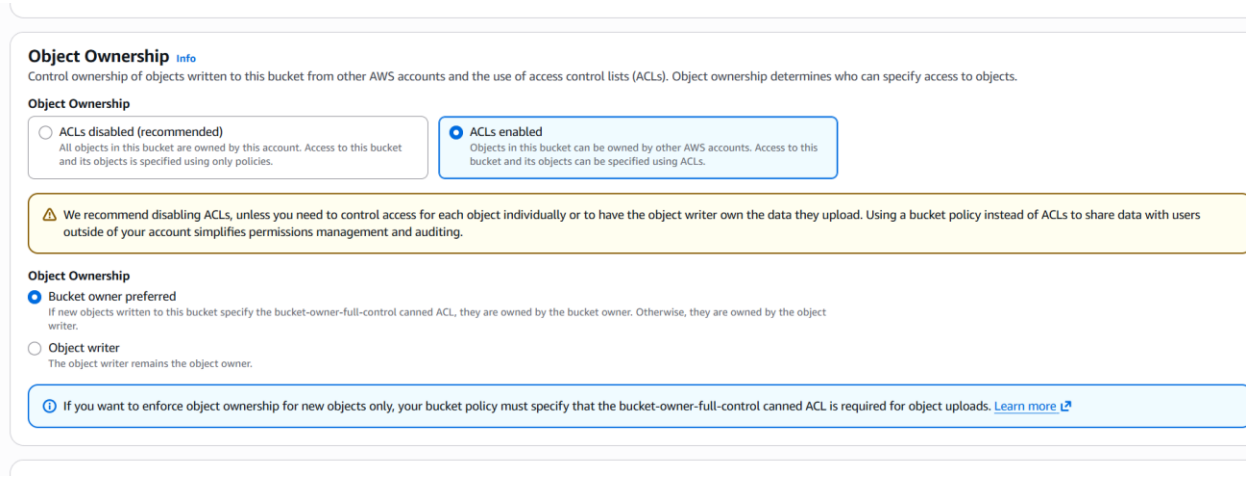
Under Block Public Access settings for this bucket:

❌ Uncheck Block all public access

You must uncheck ALL options:

✓ Check I acknowledge that this bucket will become public

👉 Click Create bucket



2⃣ Enable Public Access at Bucket Level (ACL)

Step 4: Open the Bucket

- Click on the bucket name
- Go to Permissions tab

Step 5: Edit Bucket ACL

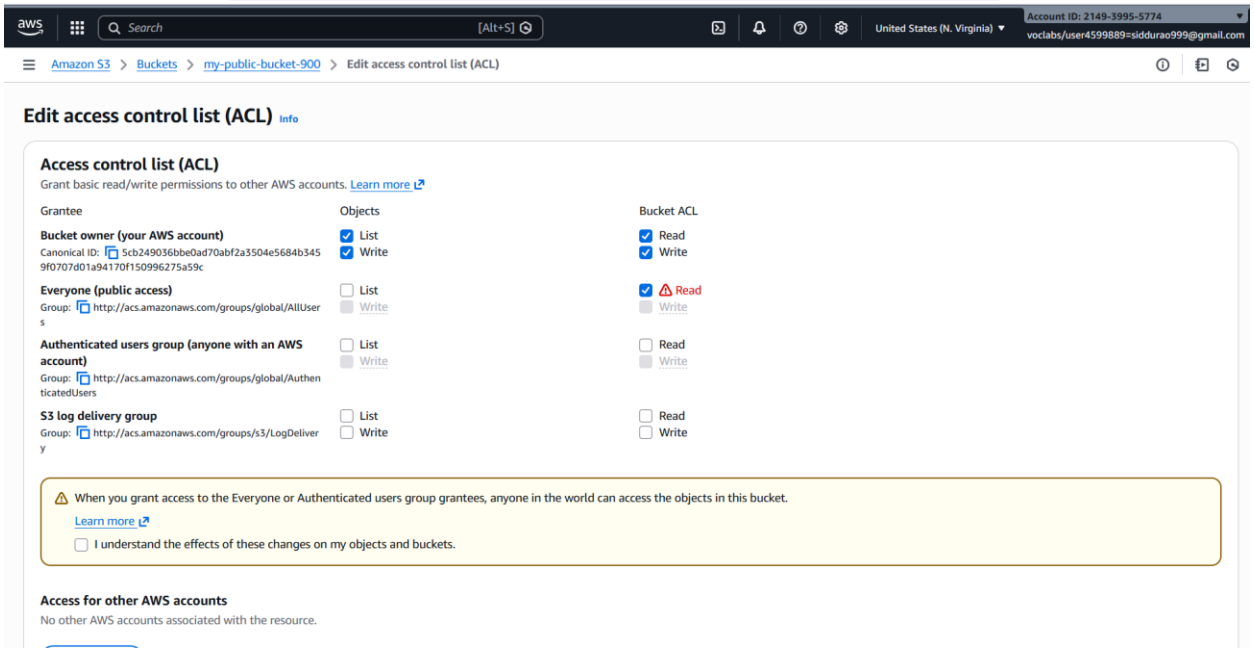
- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✓ Read access

✓ Save changes

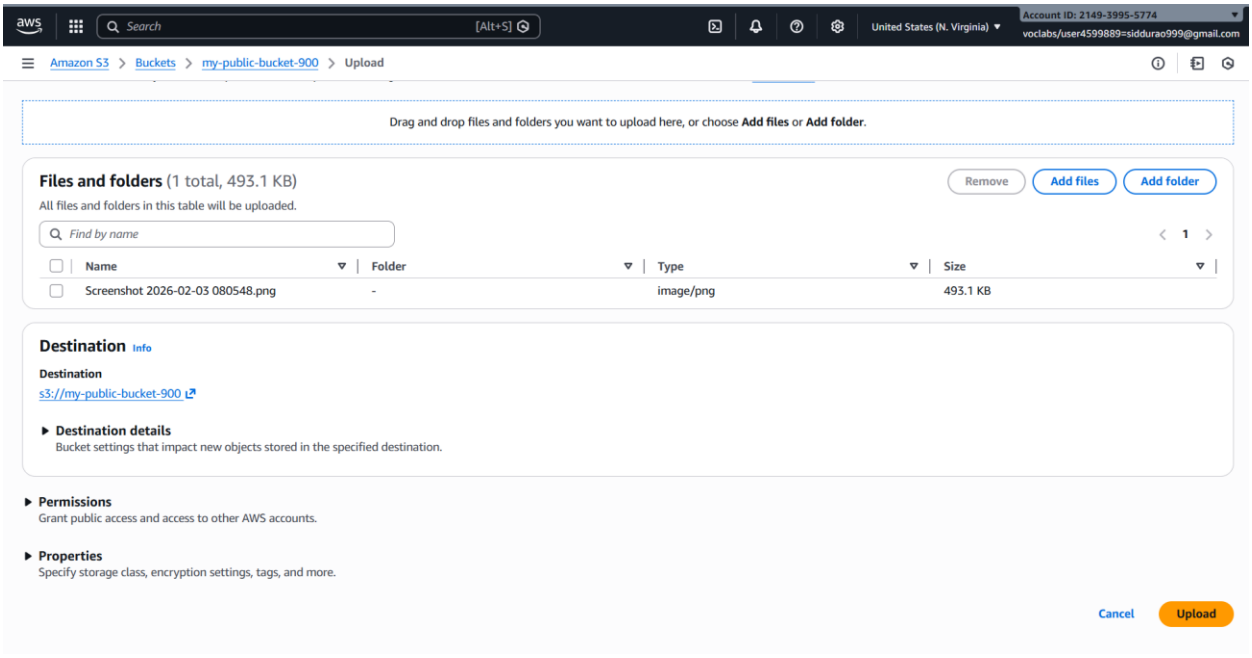
📌 **This allows public access at bucket level**

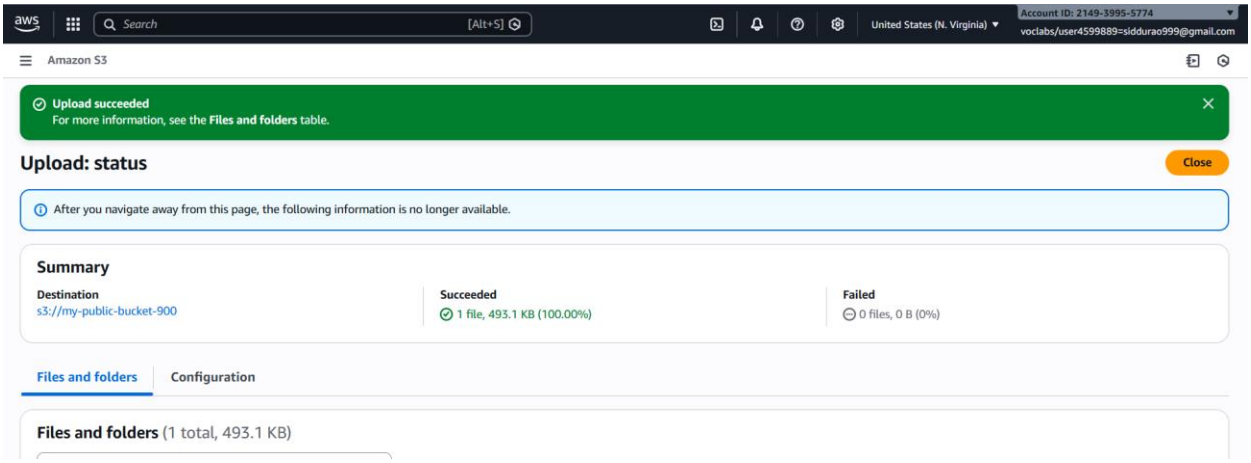


1⃣ Upload Object to the Bucket

Step 6: Upload File

- Go to Objects tab
- Click Upload
- Click Add files
- Select file (e.g., image.jpg)
- Click Upload





4⃣ Enable Public Access for Object (ACL)

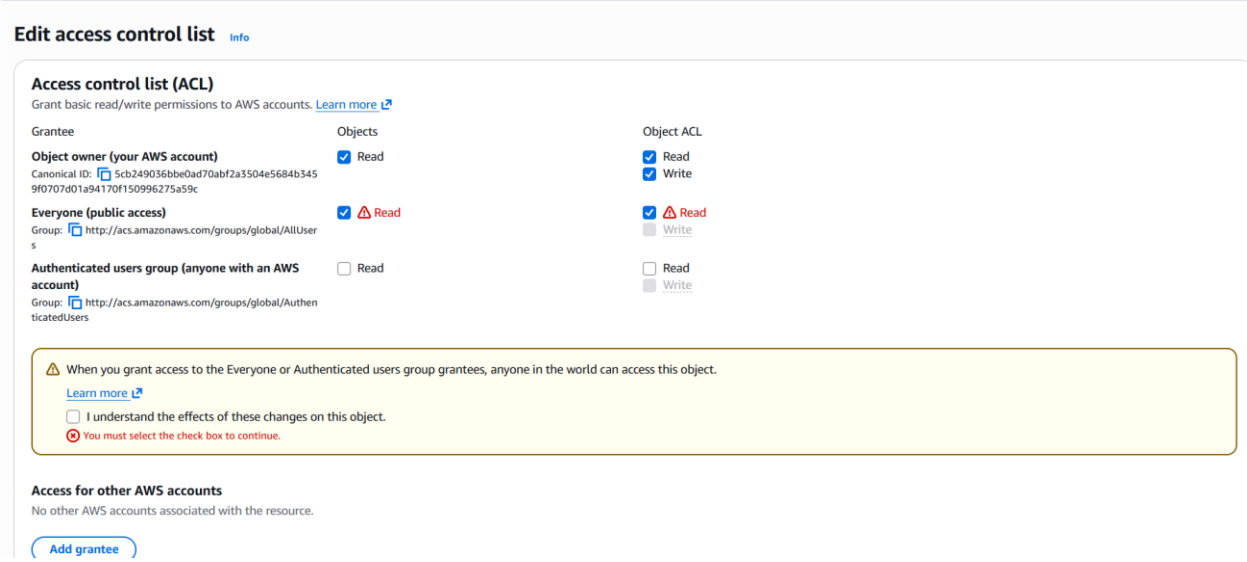
Step 7: Object-Level ACL Permission

- Click on uploaded object
- Go to Permissions tab
- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✓ Read access

✓ Save changes



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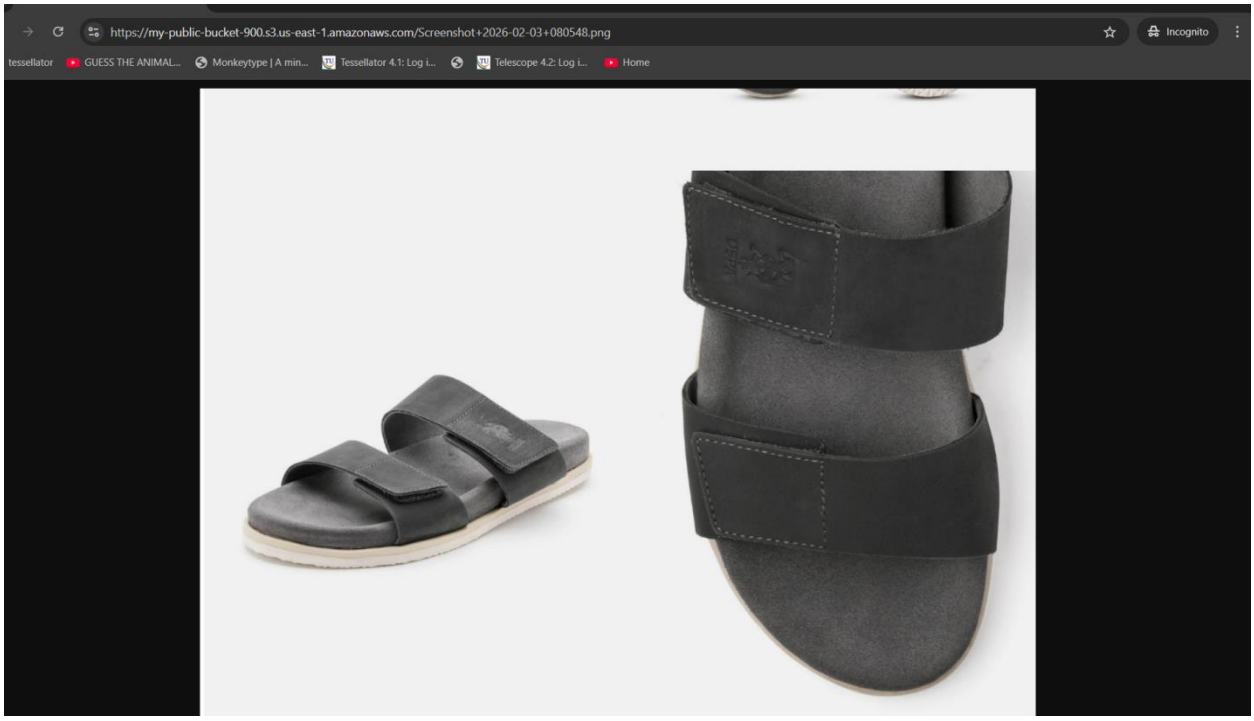
5 Verify Public Access

Step 8: Test Object URL

- Copy Object URL
- Open in browser (incognito)

✓ Object should be accessible without login



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2. S3 Versioning & Cross-Region Replication

1⃣ Enabling Versioning in S3 (with Output Scenario)

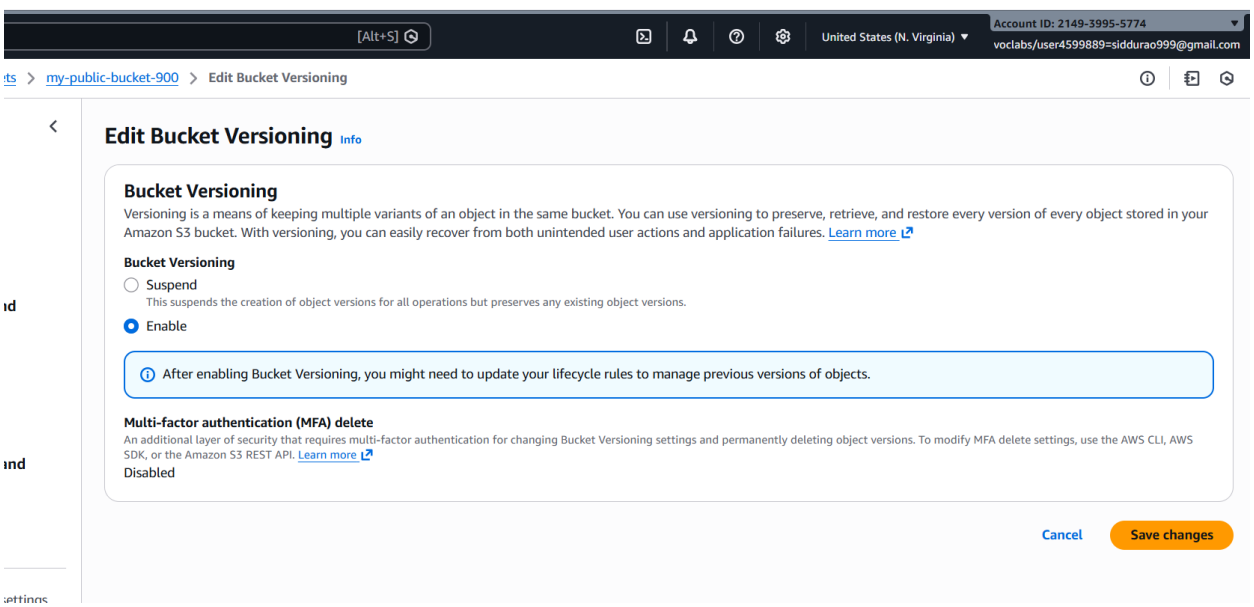
◆ Objective

To enable version control on an S3 bucket and observe object versions.

◆ Steps to Enable Versioning

1. Open AWS Management Console
2. Navigate to S3
3. Select the bucket
4. Go to Properties tab
5. Scroll to Bucket Versioning
6. Click Edit
7. Select Enable
8. Click Save changes

✓ Versioning is now enabled



◆ Output Scenario (What You Observe)

Step A: Upload Object

- Upload file: report.pdf

📌 Output:

- Version ID is assigned automatically

Step B: Modify and Re-upload Same Object

- **Upload report.pdf again (same name)**

📌 Output:

- Old version is preserved
- New version becomes current

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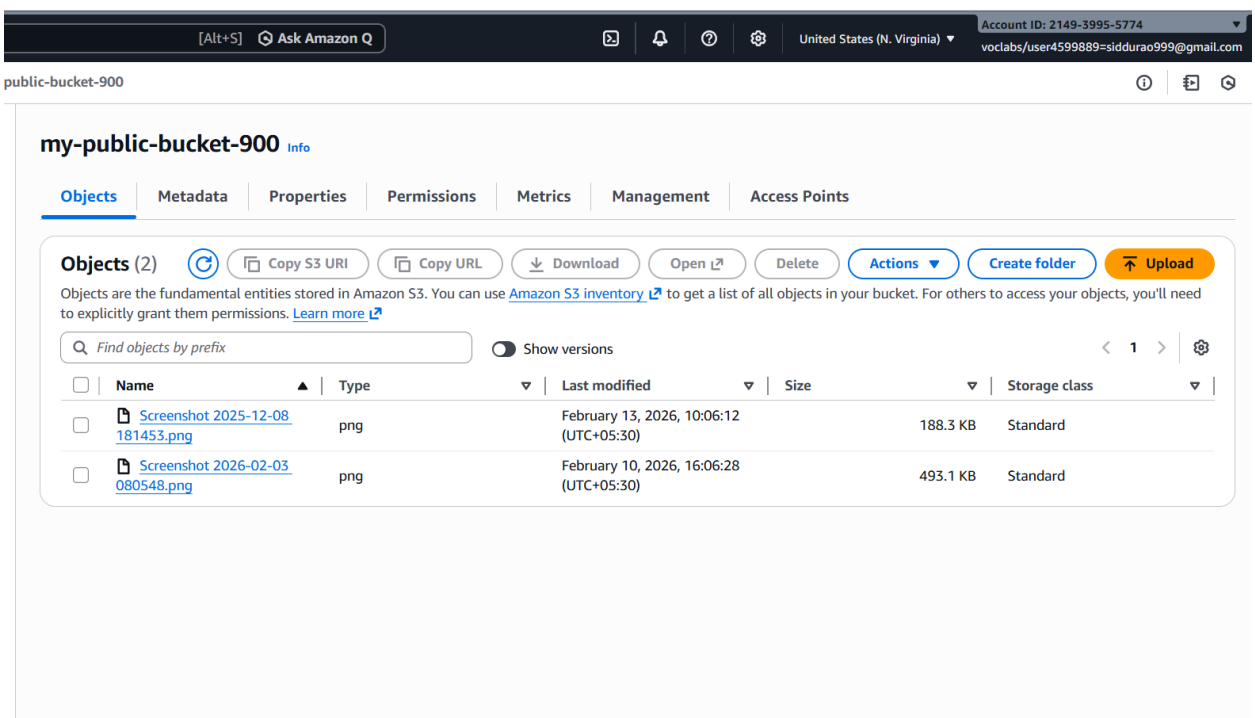
Step C: Delete Object

- Click Delete on report.pdf

📌 **Output:**

- Object appears deleted
- A Delete Marker is created
- Older versions still exist and can be restored



2 Cross-Region Replication (CRR)

◆ Objective

To replicate objects from one bucket to another in a different region using AWS-managed permissions.

◆ Prerequisites (Very Important)

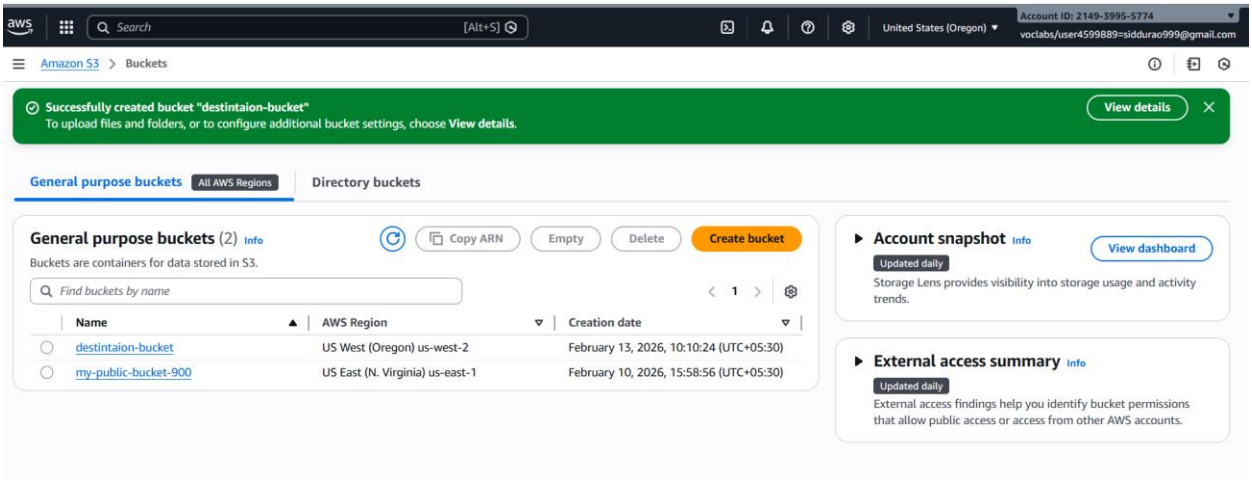
✓ Versioning must be enabled on both buckets

✓ Source and destination buckets must be in different regions

✓ AWS Academy automatically handles replication permissions

Step 1 Create Destination Bucket

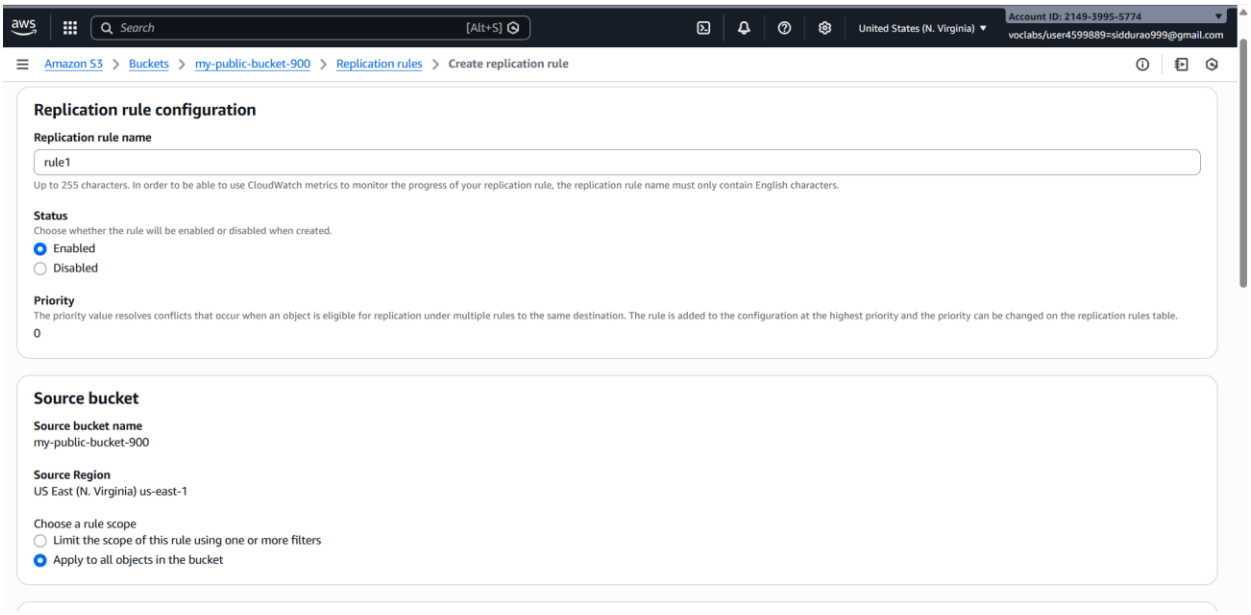
1. Go to S3 → Create bucket
2. Bucket name: dest-crr-bucket
3. Select different region
4. Complete bucket creation
5. Enable Versioning (same steps as above)



Step 2 Create Replication Rule (Without IAM Selection)

1. Open Source bucket
2. Go to Management tab
3. Click Replication rules
4. Click Create replication rule

Step 3 Configure Replication Rule

- Rule name: academy-crr-rule
- Status: Enabled
- Scope: Apply to all objects
- 

Step 4 Choose Destination Bucket

- Select Another bucket
- Choose Destination bucket
- Confirm different region

📌 **AWS Academy Note:**

You will see an option like:

"AWS will create and manage the required permissions automatically"

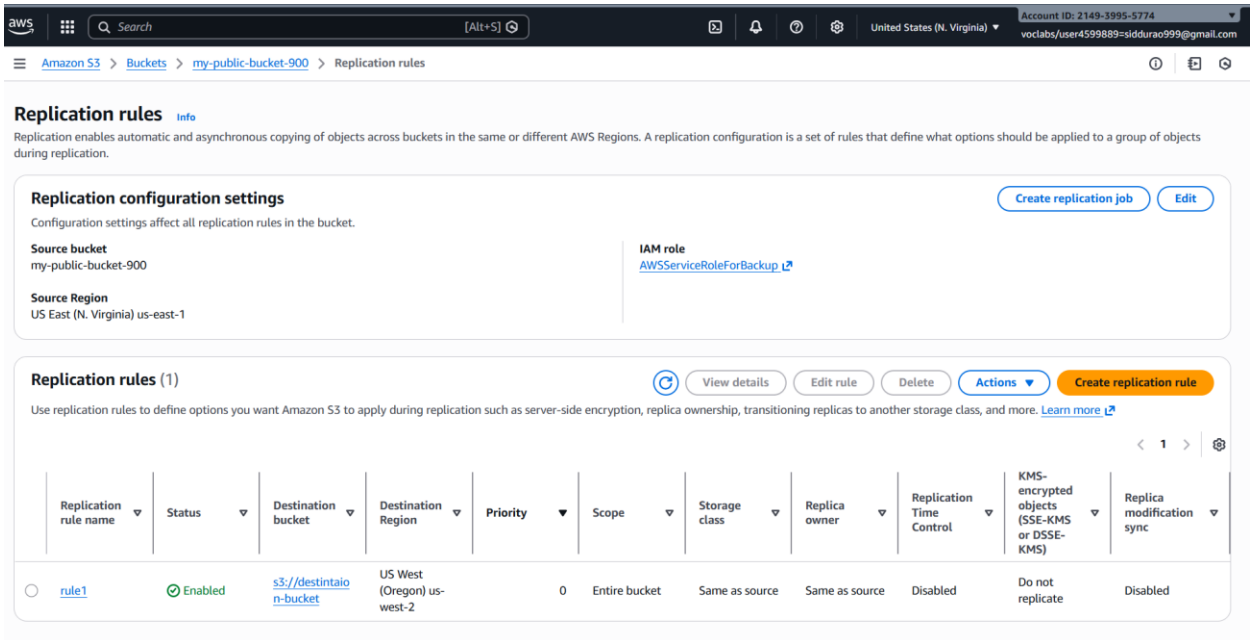
✓ Select Use AWS-managed role / default role

Step 5 Replication Options

- **Replicate:**
  - ✓ New objects
  - ✕ Existing objects (optional, depends on Academy permissions)
- **Encryption: Leave default**

Click Save

✓ **Replication rule is now active**



3️⃣ CRR Output Scenario (What You Observe)

◆ Step A: Upload New Object to Source Bucket

- Upload file: image.png

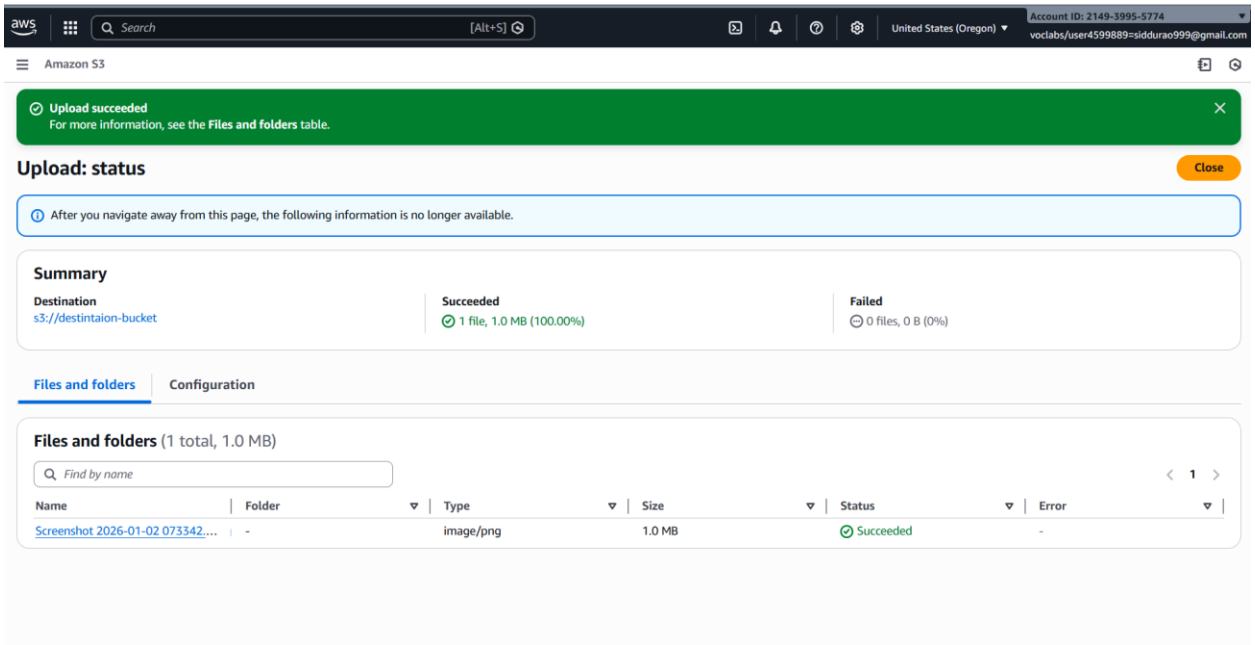
📌 Output:

- Object is uploaded normally

◆ Step B: Verify Destination Bucket

- Open destination bucket
- Object image.png appears automatically
- Version ID is present

✓ **Replication successful**



3. Static Website Hosting in S3

(Using Bucket-Level & Object-Level Permissions – ACL Method)

◈ Objective

To host a static website in Amazon S3 using index.html and error.html, and allow public access using ACL-based bucket and object permissions.

◈ Prerequisites

- AWS / AWS Academy account
- Two files:
  - index.html
  - error.html

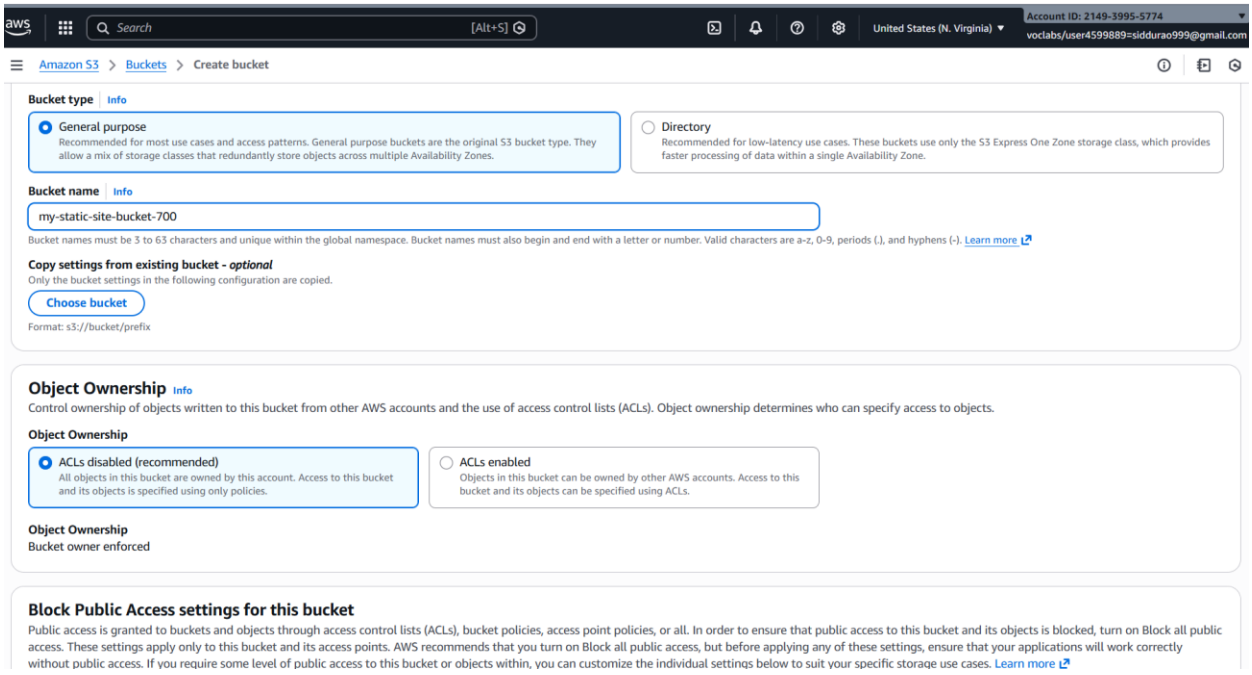
1⃣ Create an S3 Bucket

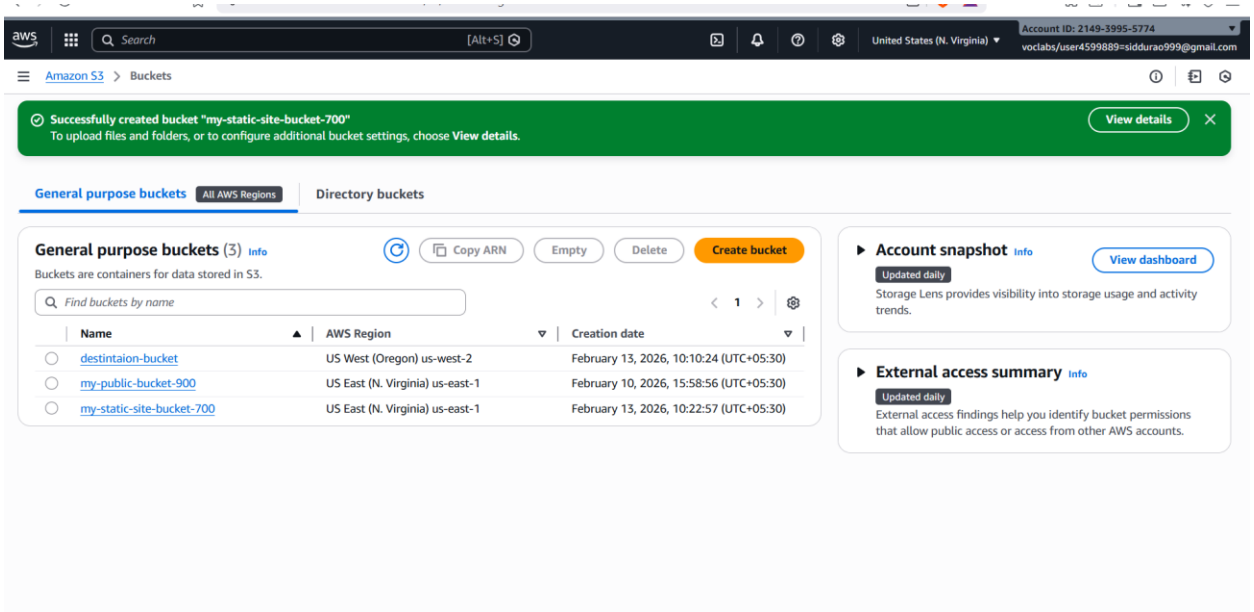
1. Login to AWS Management Console
2. Go to Services → S3
3. Click Create bucket
4. **Enter:**

- Bucket name: my-static-site-bucket-123
- Region: Nearest region

5. **Under Block Public Access:**
   - ✕ Uncheck Block all public access
   - ✓ Acknowledge the warning
6. **Click Create bucket**

✓ **Bucket created successfully**

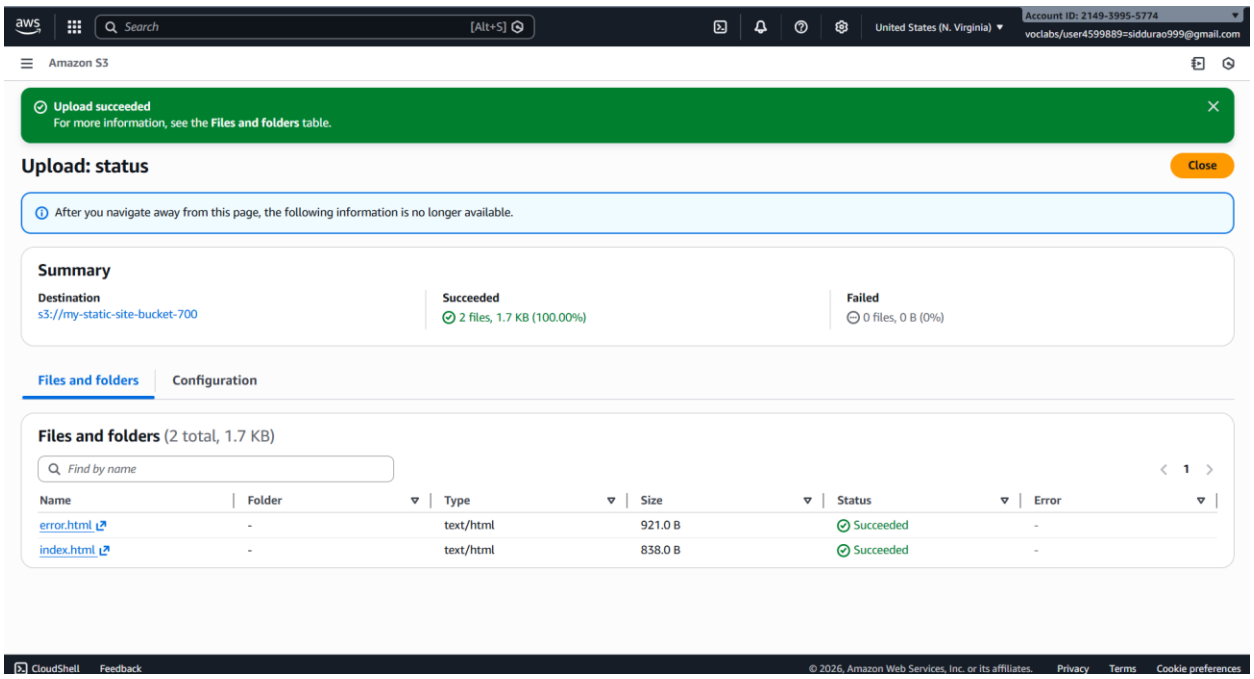




2️⃣ Upload Website Files

1. Open the bucket
2. Go to Objects tab
3. Click Upload
4. Click Add files
5. Select:
    - index.html
    - error.html
6. Click Upload

✓ **Files uploaded**



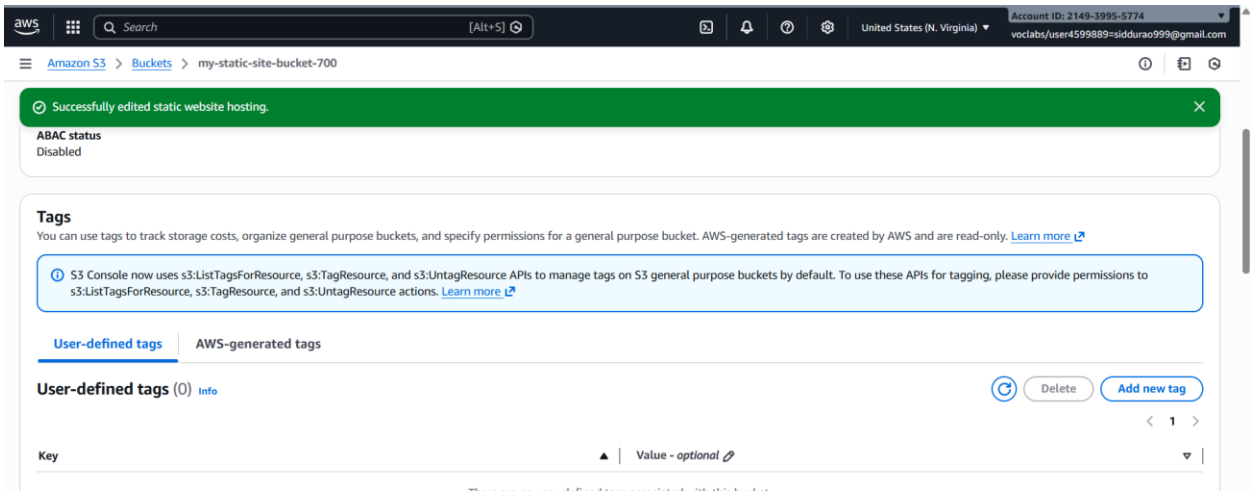
3️⃣ Enable Static Website Hosting

1. Go to Properties tab
2. Scroll to Static website hosting
3. Click Edit
4. Select Enable
5. Choose Host a static website
6. Enter:

- Index document: index.html
- Error document: error.html

7. **Click Save changes**

✓ **Static website hosting enabled**



4⃣ Enable Bucket-Level Public Access (ACL)

◈ Purpose

Allows public users to access objects inside the bucket.

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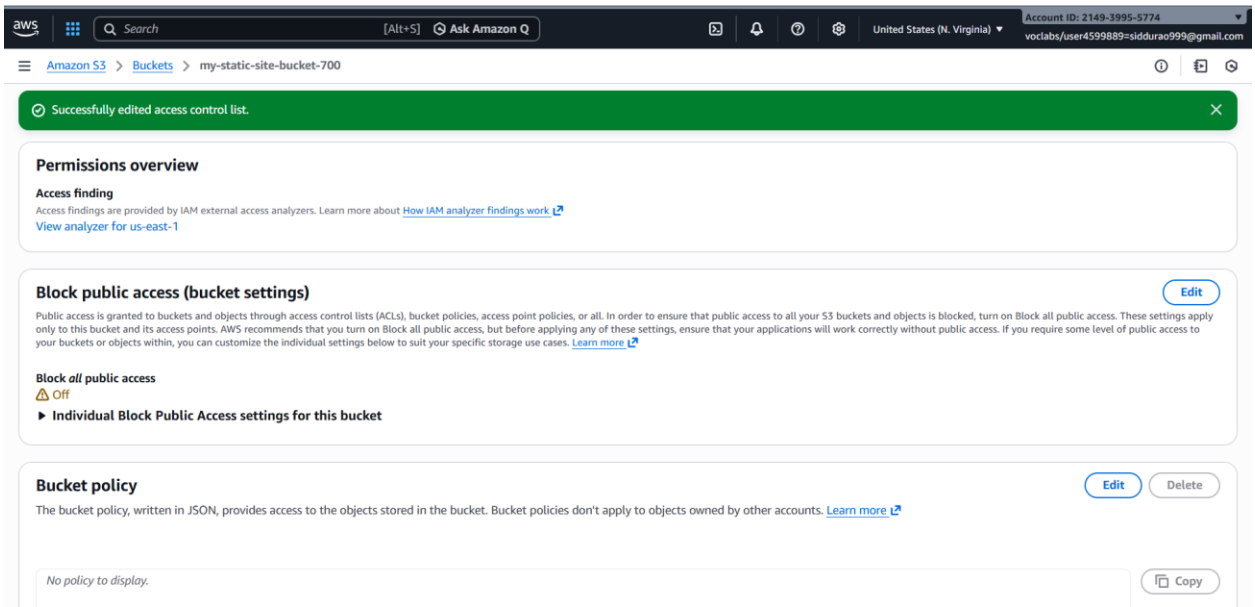
1. Go to Permissions tab
2. Scroll to Access Control List (ACL)
3. Click Edit
4. Under Public access:
   - Everyone (public access) → ✓ Read
5. Click Save changes

✓ Bucket-level public read access enabled



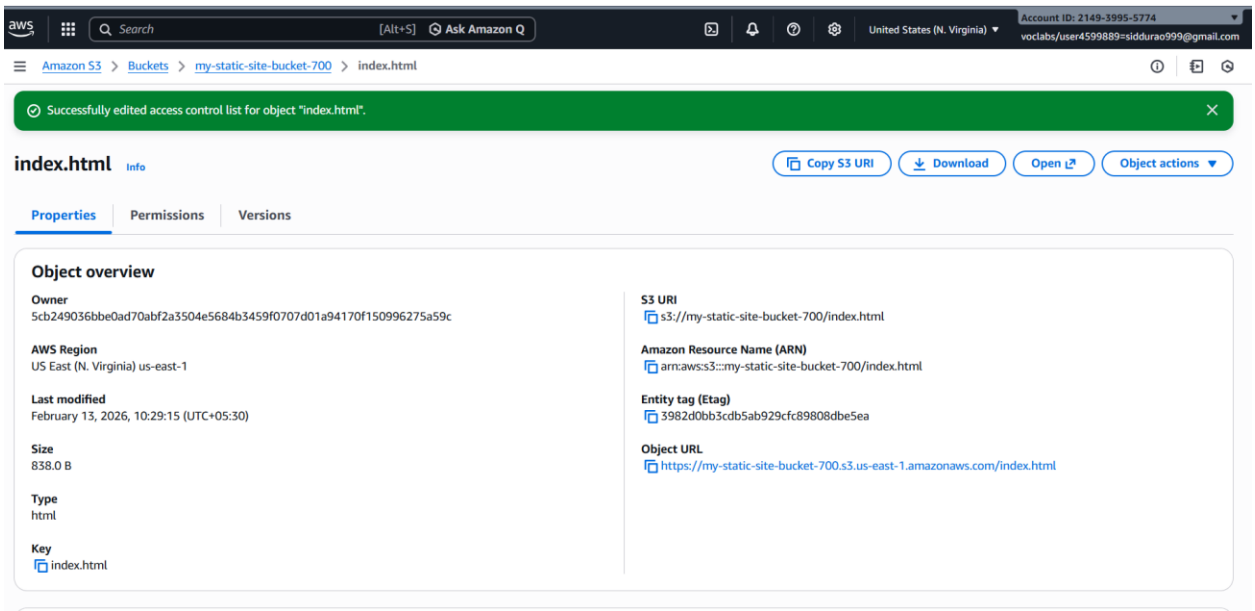
5⃣ Enable Object-Level Public Access (ACL)

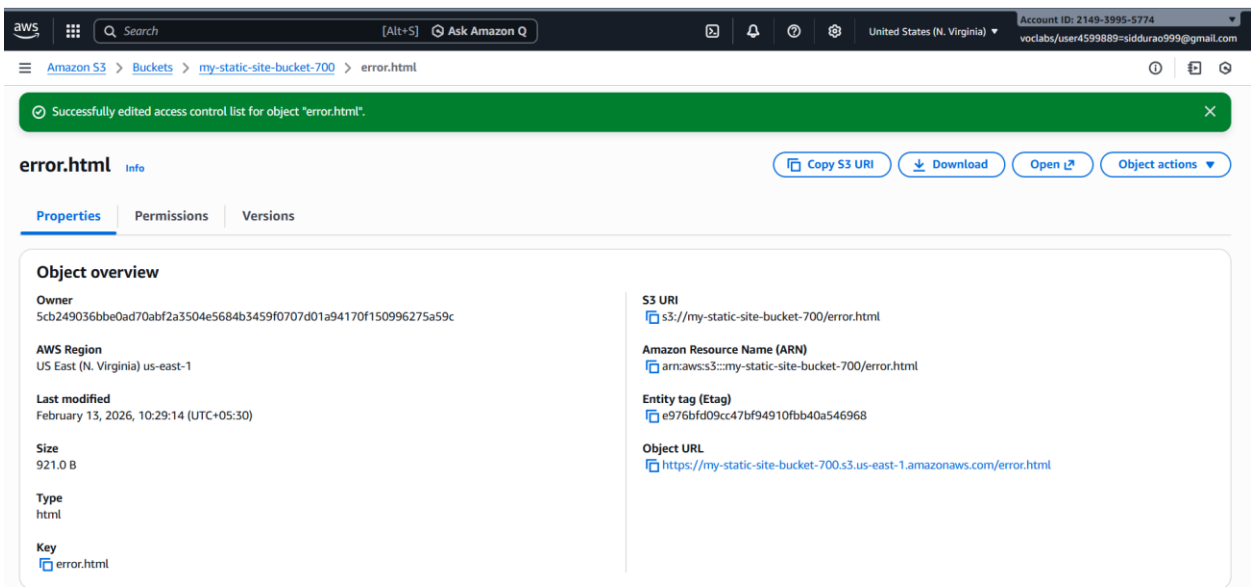
Steps (For Each File)

1. Go to Objects tab
2. Click on index.html
3. Go to Permissions
4. Scroll to Access Control List (ACL)
5. Click Edit
6. Under Public access:
   - Everyone (public access) → ✓ Read
7. Save changes

🔁 **Repeat the same steps for error.html**

✓ **Object-level public access enabled**

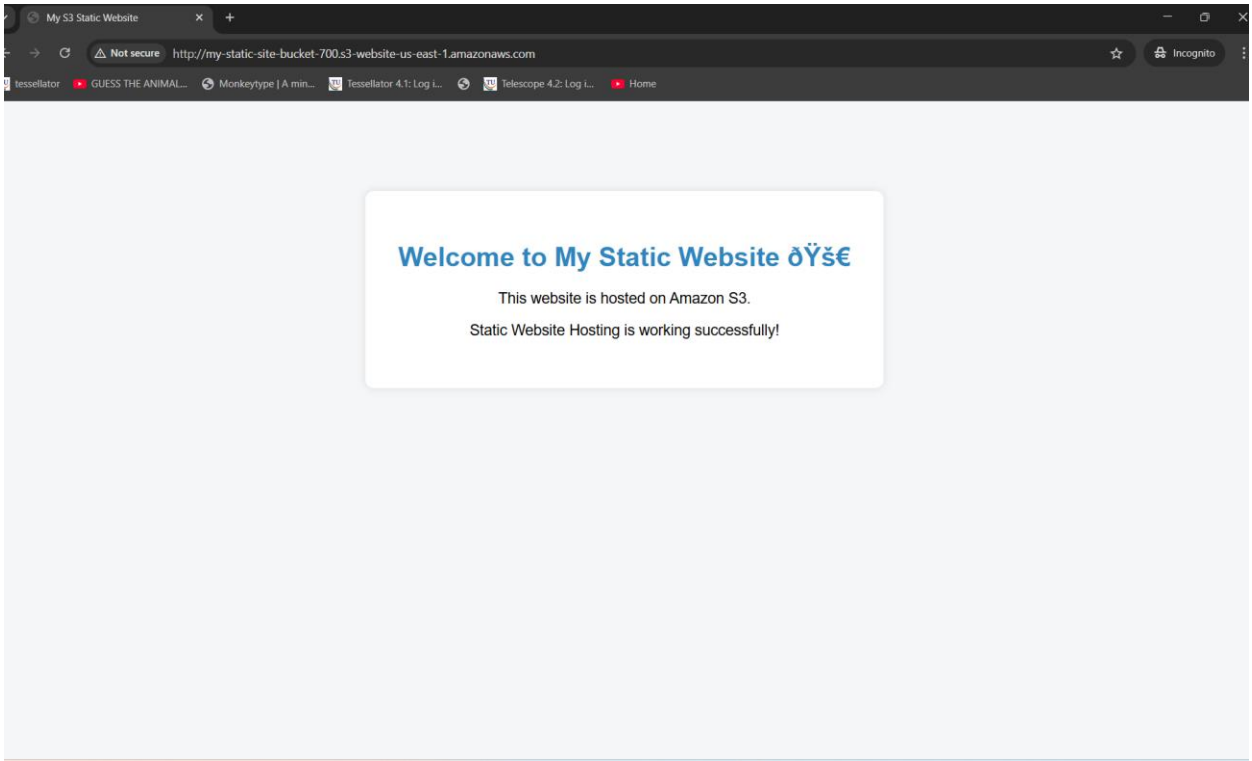




6.Access the Static Website

1. Go to Properties
2. Scroll to Static website hosting
3. Copy the Bucket website endpoint
4. http://my-static-site-bucket-123.s3-website-region.amazonaws.com
5. Paste into browser

✓ index.html page loads successfully



7️⃣ Error Page Output Scenario

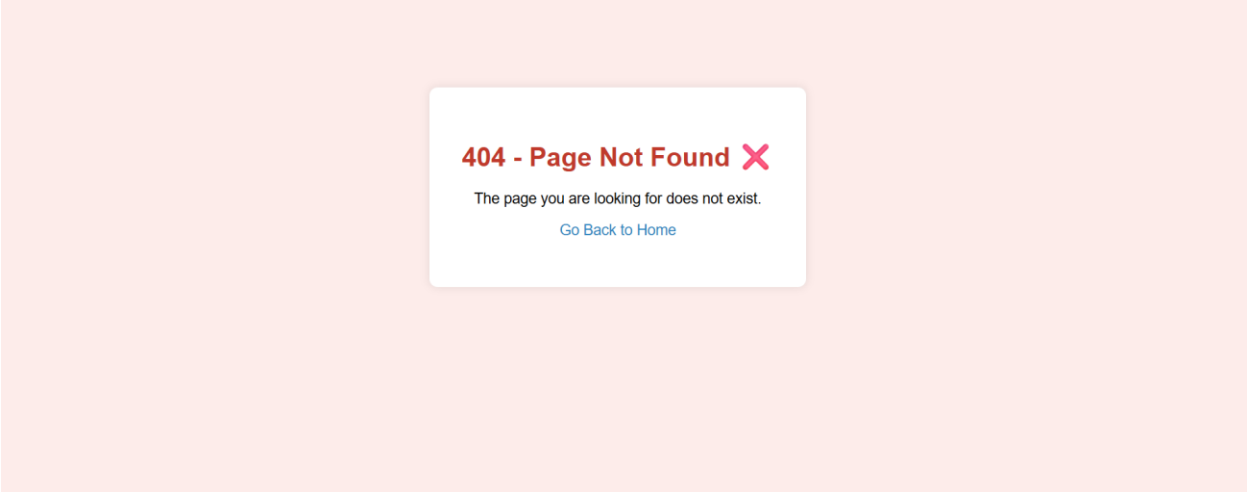
🔷 Test Error Page

- Open an invalid URL:
- http://bucket-name.s3-website-region.amazonaws.com/invalid.html

📌 Output:

- error.html page is displayed

✅ Error page working correctly



Amazon S3 – Scenario-Based Questions

1. Object uploaded but not accessible from browser. What permission issue could be the cause?
Most likely, public access is not enabled.
Either Block Public Access is still ON, or the object does not have public read permission.

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2. Bucket policy allows public read, but one object is still not accessible. Why?
Because object-level ACL does not allow public read.
Even if the bucket policy is public, the object itself must also allow read access.

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3. No bucket policy, but objects have public ACLs. Can users access them?
Yes, users can access those objects.
The final access decision depends on a combination of bucket policy, object ACL, and block public access settings.

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4. How do you ensure no public access even if someone applies a public ACL by mistake?
Enable Block Public Access at the bucket level.
This overrides all public bucket policies and ACLs.

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5. Bucket creation failed due to naming error. What rule was violated?
Possible violations:

- **Bucket name is not globally unique**
- **Contains uppercase letters**
- **Contains special characters**
- **Name is too short or too long**

S3 bucket names must follow strict naming rules.

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6. Two AWS accounts try to create the same bucket name. One fails. Why?
Because S3 bucket names are globally unique across all AWS accounts.
Only one account can own a bucket name.

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7. Static website enabled but URL returns 403 Forbidden. Why?
Public access is not configured properly.
Either:

- **Block Public Access is enabled**
- **Bucket or objects do not have public read permission**

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8. Static website enabled but root URL says index.html not found. What's the issue?

The index.html file is missing, wrongly named, or not uploaded.
S3 is trying to load index.html but cannot find it.

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9. What HTTP error appears if index document is missing and why?
S3 returns a 403 Forbidden error.
This happens because S3 does not allow directory listing like a web server.

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10. Bucket policy allows public access, but website still doesn't work. What else to check?
Check object-level ACLs.
Each website file like index.html must also have public read access.

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11. You overwrote an important object, but versioning was enabled. How do you recover it?
I can go to Object versions, find the older version, and either restore or download it.
Versioning keeps all previous copies.

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12. A user deletes an object from a versioned bucket. Is the data lost?
No, the data is not permanently deleted.
S3 creates a delete marker, and older versions still exist and can be restored.

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13. CRR is enabled but objects are not replicating. What might be missing?
Most likely:

- **Versioning is not enabled on one of the buckets**
- **Or the replication role/permissions are missing**

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14. Only new objects replicate after enabling CRR. Why not old ones?
Because CRR works only for objects uploaded after replication is enabled.
Old objects need manual replication or a separate option.

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15. What are the key prerequisites for enabling Cross-Region Replication?

- **Versioning must be enabled on both buckets**
- **Buckets must be in different regions**
- **Proper IAM permissions or AWS-managed role**
- **Destination bucket must exist**